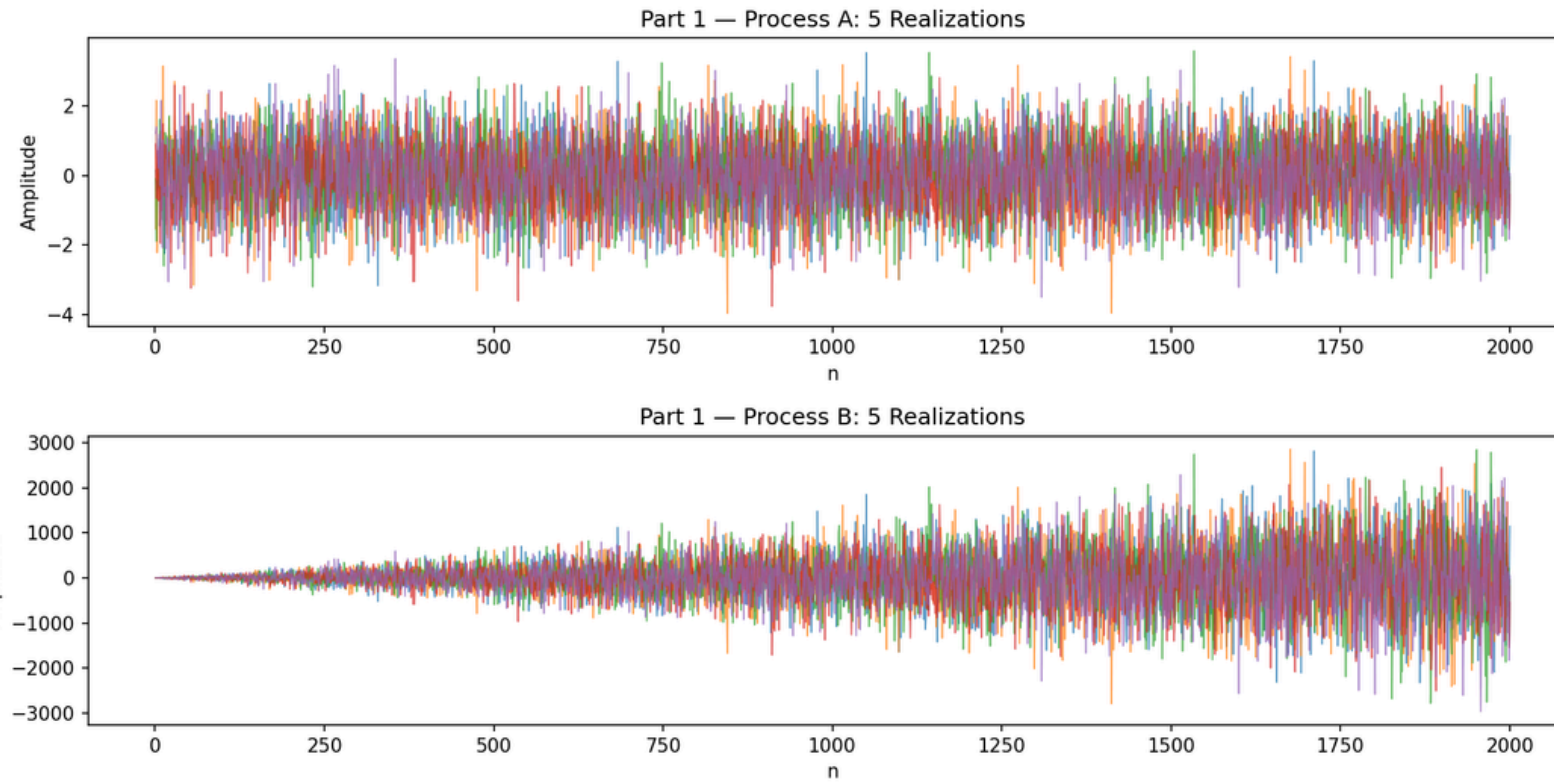
name :	hossamEldenMohamed Ahmed Mohamed
Id:	9230331

Answers

Part 1:

Task 1:



Explanining:

5 realizations of Process A (white noise, constant amplitude spread) and Process B (amplitude grows with n due to the (1+0.5n) scaling).

Part 1 Task 2: Ensemble Mean:

n	mean_A	mean_B
100	0.00556	0.284
1000	−0.00286	−1.434
2000	−0.00326	−3.267

explaining:

Process A mean ≈ 0 at all points. Process B mean drifts and scales with n.

E[X_B[n]] = (1+0.5n)·E[W[n]] = 0 theoretically since E[W]=0. The observed values appear to grow because the estimator variance is $(1+0.5n)^2/K$, so the statistical noise of the estimate itself scales with n. This is not a nonzero mean it is evidence that the variance of X_B grows with n, which confirms non-stationarity.

Part 1 Task 3: Ensemble Autocorrelation

(n1, n2)	R_A	R_B
(100, 150)	0.0005	1.765
(1500, 1550)	−0.0020	−1179.57

Explaining

R_A ≈ 0 for both pairs since n1 ≠ n2 and white noise samples are uncorrelated. Theoretically R_B should also be 0 for n1 ≠ n2, but the estimator noise has variance proportional to $(1+0.5\cdot n1)^2(1+0.5\cdot n2)^2$. At n=100–150 the scale factors are small so the noise is ~1.765, but at n=1500–1550 the scale factors are ~(751)(776) making the noise blow up to ~1179. Both pairs have the same lag τ=50, yet the autocorrelation estimate changes by a factor of ~668x this cannot happen in a WSS process where R depends only on lag. This directly proves Process B is not WSS.

Part 1 Task 4: WSS Decision

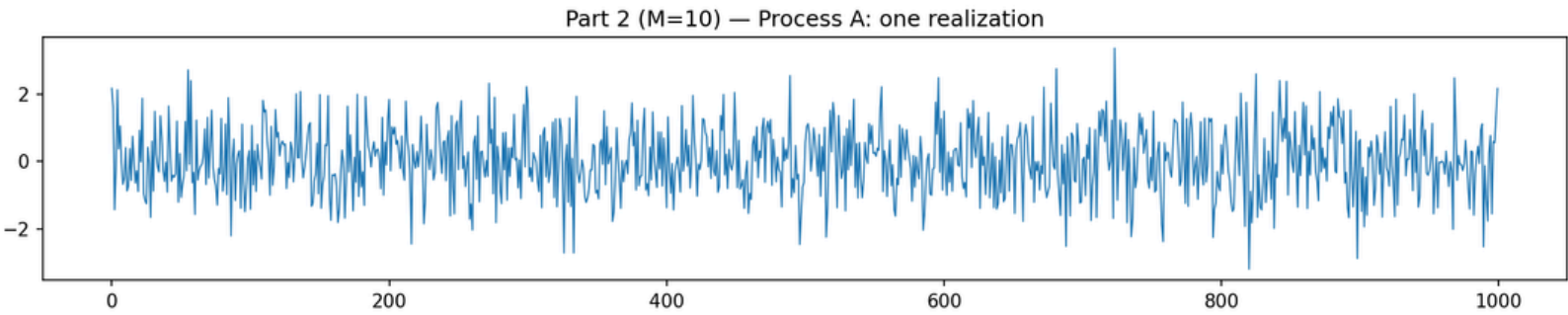
Process A: WSS. Mean = 0 (constant). $R_A(n1,n2) \approx 0$ whenever $n1 \neq n2$, depends only on lag $\tau = n2-n1$.

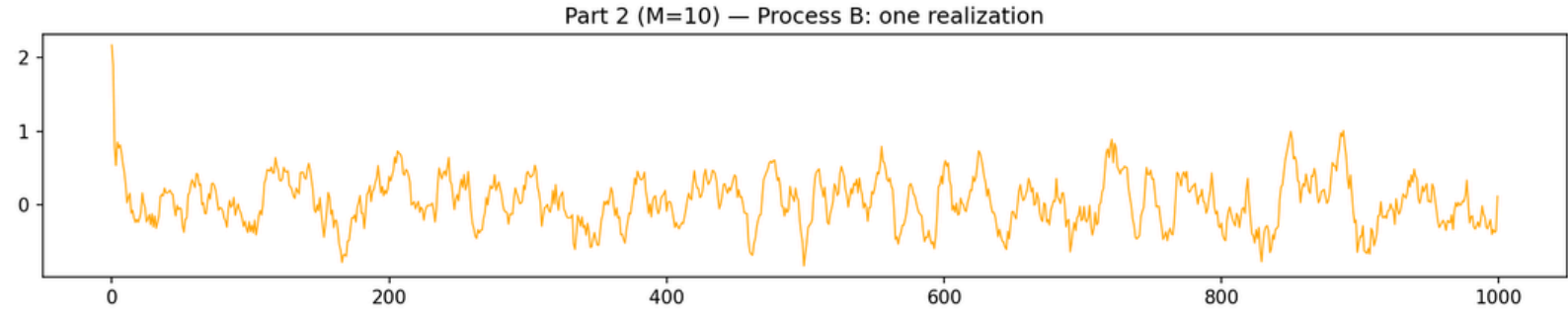
Process B: Not WSS. Mean = 0 theoretically, but variance grows with n.. The autocorrelation $R_B(n1,n2) = (1+0.5n1)(1+0.5n2)\cdot\delta(n1-n2)$,

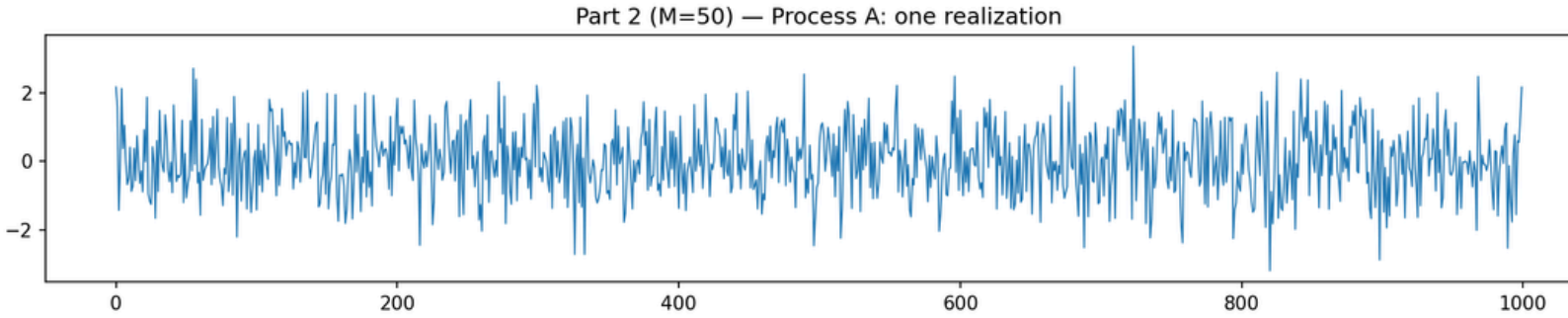
which depends on the absolute values of n1,n2, not just the lag. The variance $(1+0.5n)^2$ is unbounded.

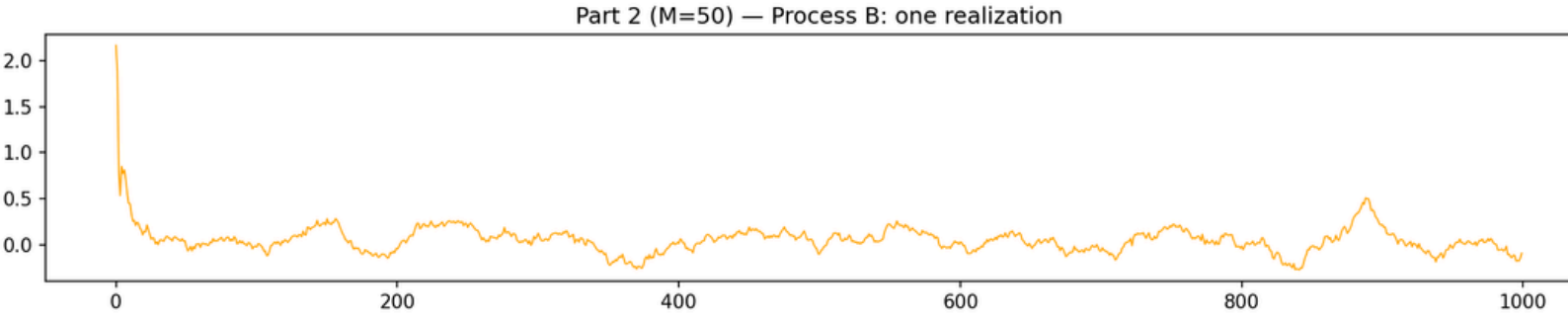
At $\tau=0$: $R_B(n,n) = (1+0.5n)^2$, which grows unboundedly with n. WSS requires $R(\tau=0)$ to be a finite constant, so this alone is sufficient to conclude Process B is not WSS.

Part 2 Task 1



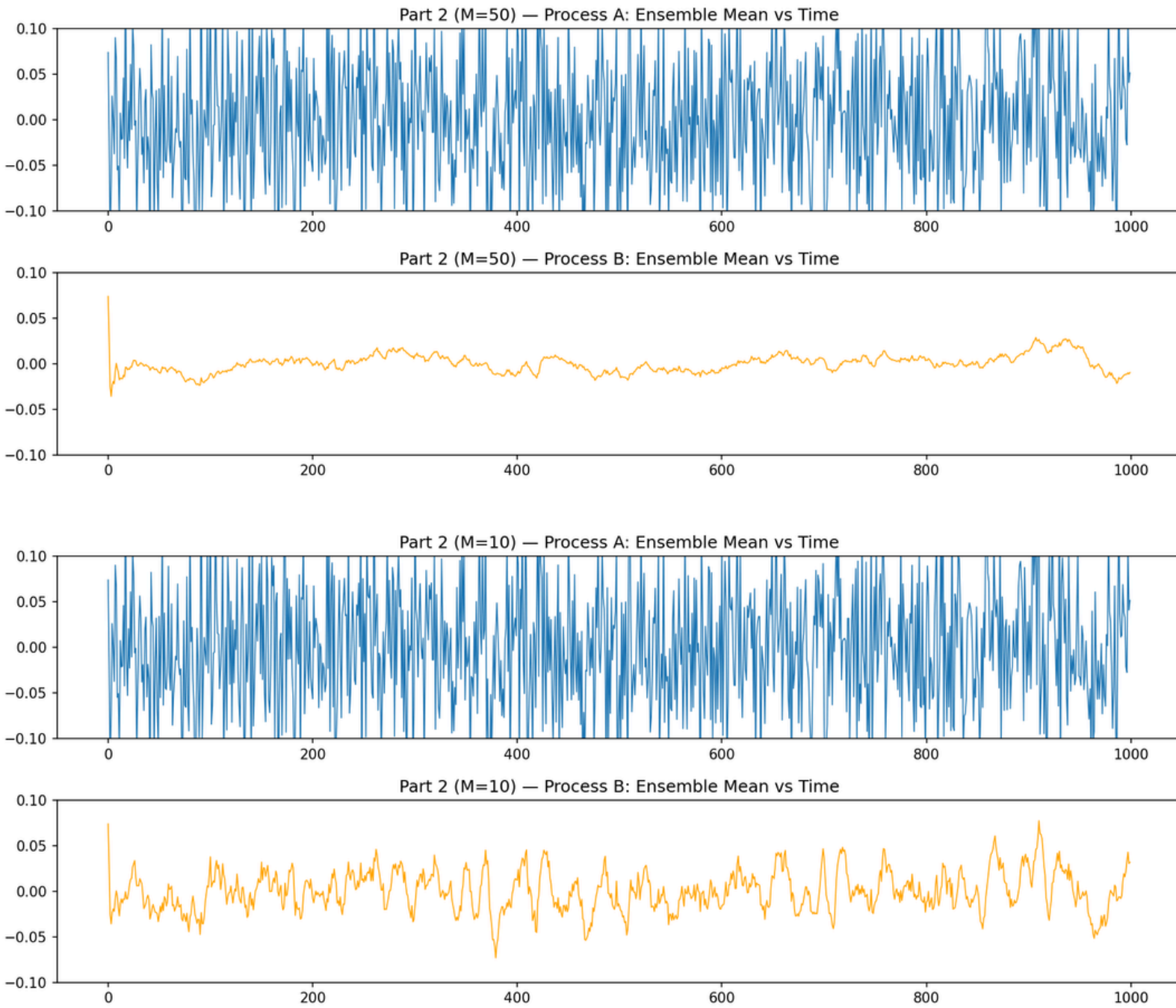






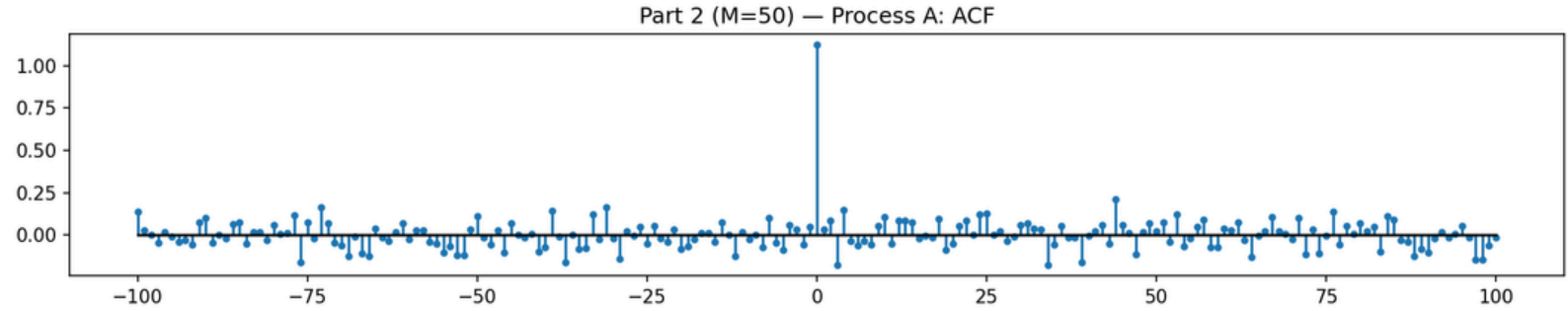
Process A varies rapidly sample-to-sample (white noise). Process B varies slowly the moving average smooths out fast fluctuations.

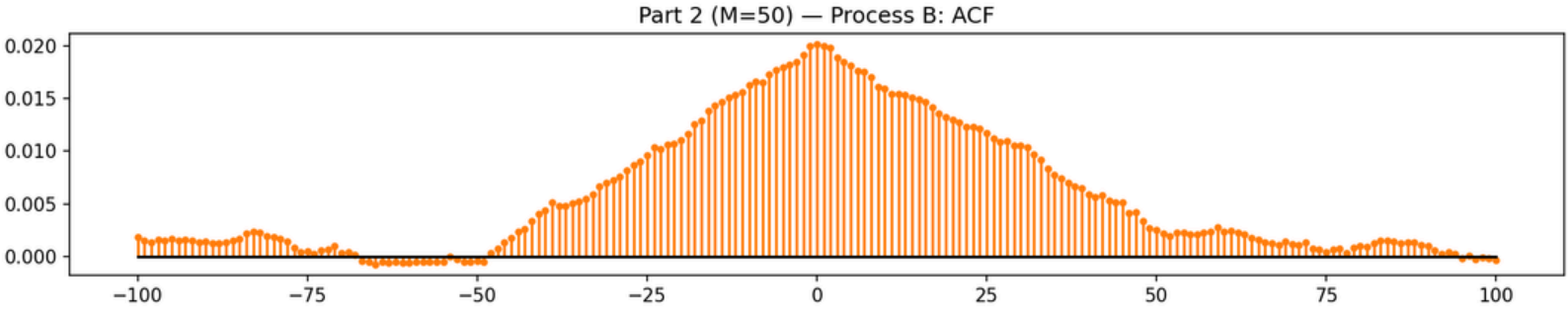
Part 2 Task 2: Ensemble Mean vs Time

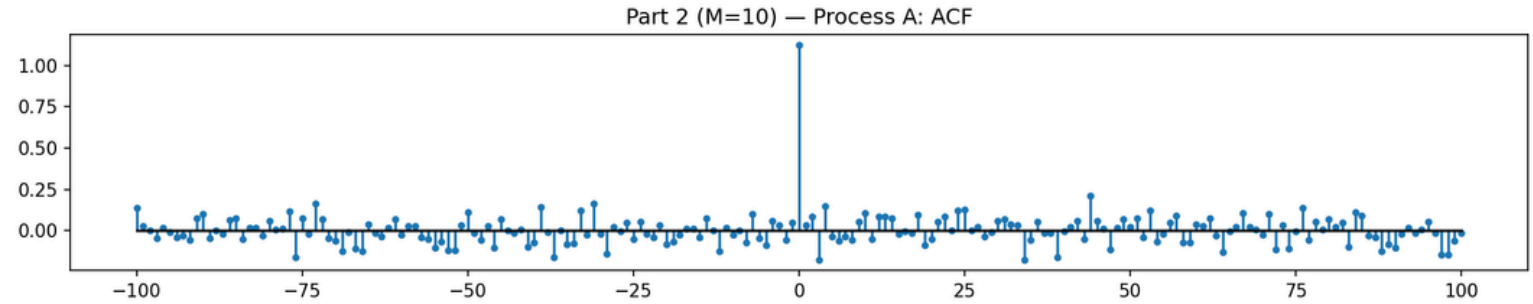


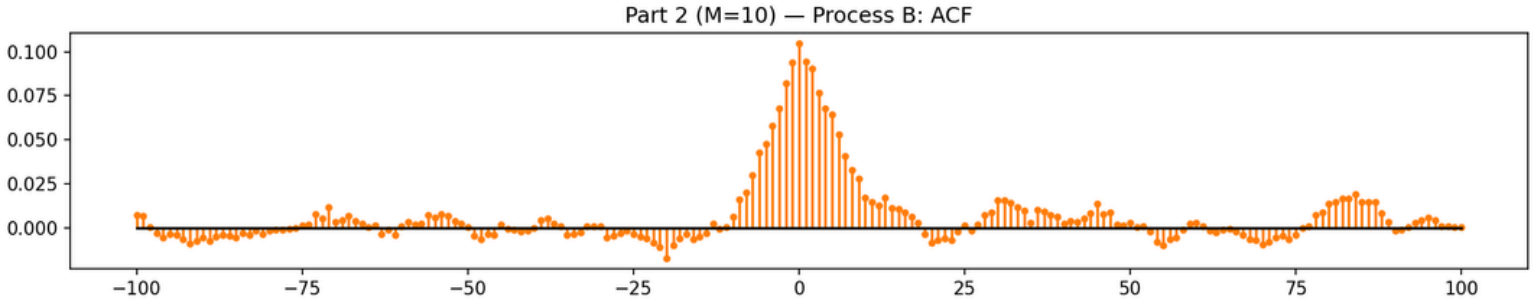
Both processes have ensemble mean ≈ 0 for all n, confirming WSS condition 1. The visible fluctuations in the plot are purely statistical noise from using only K=200 realizations the standard error is 1/√200 ≈ 0.07, which matches the ±0.1 range seen in the plots.

Part 2 Task 3: ACF



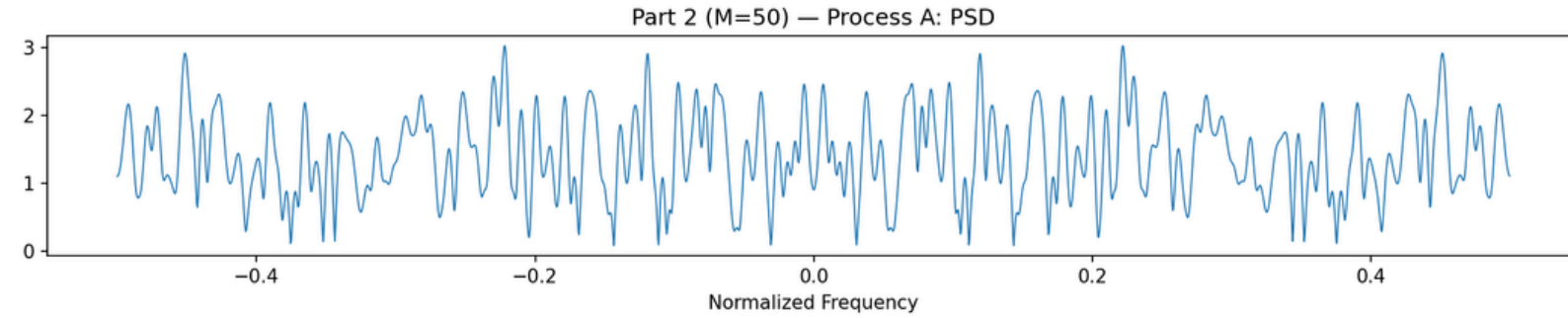


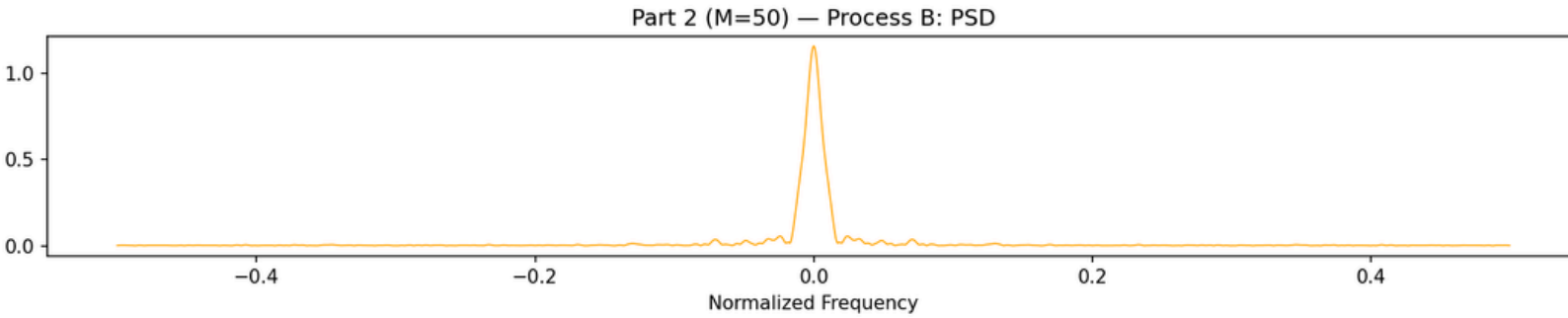


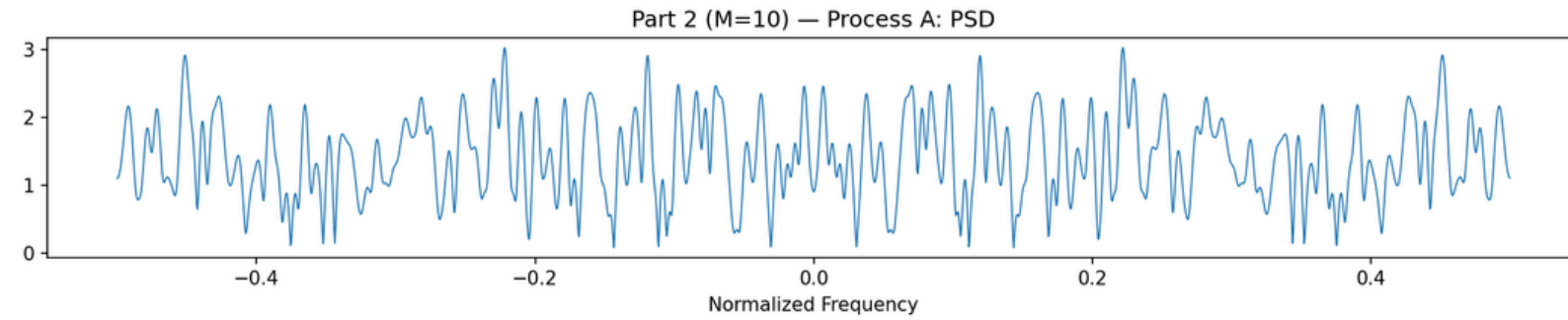


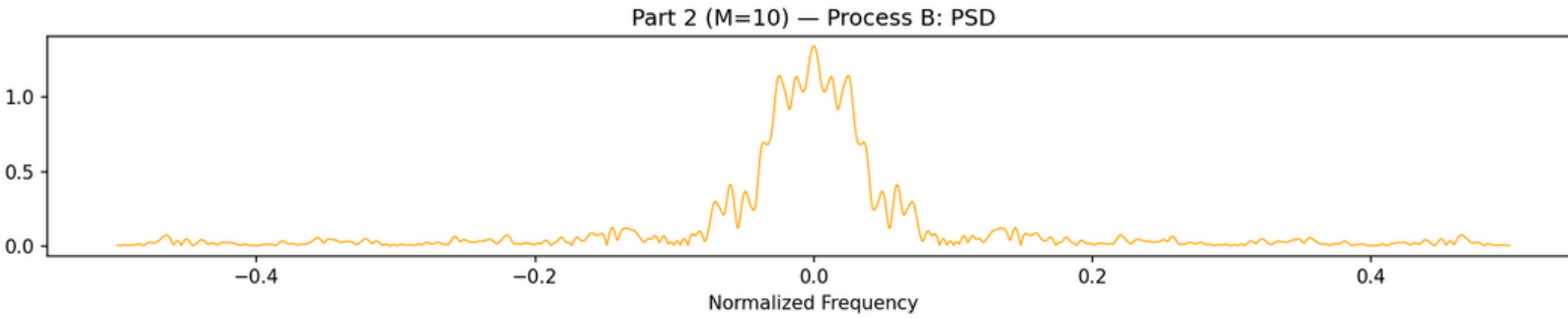
Process A: impulse at τ=0, zero elsewhere. Process B M=10: nonzero over roughly |τ| ≤ 10. Process B M=50: nonzero over roughly |τ| ≤ 50.

Part 2 Task 4: PSD









Process A PSD appears noisy but is fundamentally flat (white spectrum). The roughness is due to spectral leakage from the short ACF window of only 201 points (τ from −100 to 100) a short window in the ACF domain means poor frequency resolution after FFT. Process B shows a clear lowpass shape with bandwidth ≈ 1/M, confirming the moving average acts as a lowpass filter.

Part 2 Task 5: Relation between ACF width, PSD width, and time variation

Wider ACF ↔ narrower PSD ↔ slower time variation. This is the Wiener-Khinchin duality: the PSD is the Fourier transform of the ACF. A wider ACF means more correlation between samples spaced far apart, which means the signal changes slowly. A narrower PSD means less high-frequency content, confirming slow variation.

Part 2 Task 6: Effect of M

Increasing M from 10 to 50 averages over more samples, which: makes time variation slower, widens the ACF (nonzero over $|\tau| \leq M$), and narrows the PSD bandwidth ($\approx 1/M$). The three effects are directly linked M controls the trade-off between time resolution and frequency resolution.